

Multilingual SMS Spam Detection for English and Bangla Using BiLSTM and Fine-Tuned XLM-R with LIME-Based Explainability

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Section: B

Abstract—Unsolicited and fraudulent SMS messages continue to be an issue. Spam filters that only work for text do not work well in places where people mix English and Bangla. This paper looks at detecting spam in SMS messages in a setting. The study used a collection of 5,633 messages. Out of these 5,158 were in English. 475 Were in Bangla. 15% Of all messages were spam. The data was divided into 4,506 messages for training and 1,127 for testing. The study compared two models. One was a Bidirectional LSTM model, called BiLSTM. The other was a model called XLM-RoBERTa which was fine-tuned. The BiLSTM used a Keras tokenizer with a vocabulary of 10,000 words. The XLM-R model had 12 transformer layers and a hidden size of 768. The study also used LIME to explain the predictions made by the models. On the test set XLM-R achieved 98.14% accuracy, 92.49% precision 95.24% recall and a 93.84% F1-score. The BiLSTM model got 97.78%, 96.13%, 88.69% and 92.26% respectively. XLM-R reduced the number of missed spam messages from 19 to 8. However it also led to false alarms going from 6 to 13. The model performed better for English messages with 98.94% accuracy and 95.79% F1-score. For Bangla messages the accuracy was 88.37% and F1-score was 87.50%. However the recall for Bangla messages was still 92.11%. LIME analysis showed that phone numbers, campaign identifiers, price expressions and reward-related words were the factors that influenced spam predictions. The results show that multilingual transformer models are good for filtering SMS messages in two languages. The main need for improvement is Bangla data.

Index Terms—SMS spam detection, Bangla, NLP, BiLSTM, XLM-RoBERTa, LIME, explainable AI

I. INTRODUCTION

SMS remains a communication channel for personal messages, financial alerts, and one-time passwords, but it is also used for advertising, phishing, and fraud. Short messages offer little context and often contain abbreviations, spelling variation, phone numbers, prices, and service codes. These properties make classification difficult. In Bangladesh and other Bangla-speaking settings, messages may use Bangla script, English, or a mixture of both, while many established SMS benchmarks are English-language collections.

This study asks whether a multilingual transformer can improve bilingual SMS spam classification over a recurrent neural-network baseline, and what kinds of message features contribute to its predictions. The work compares BiLSTM and fine-tuned XLM-R on a shared train/test split, reports aggregate and per-language performance, and applies LIME to selected messages. Its contributions are the combined English–

Bangla evaluation, separate language-level reporting, and an example-based inspection of model explanations. These findings are exploratory: the Bangla sample and especially its test subset are small, and the reported experiment uses a single split.

II. RELATED WORKS

The SMS Spam Collection introduced by Almeida et al. became a widely used English benchmark for spam filtering [1]. Subsequent studies compared traditional machine-learning approaches with learned representations. Roy et al. evaluated deep-learning and conventional models for SMS filtering [2], while Abayomi-Alli et al. studied deep learning on locally collected SMS data [3]. These works motivate recurrent baselines but do not provide the same English–Bangla comparison as the present study.

Long short-term memory networks were designed to model sequential dependencies [4]; bidirectional recurrent networks incorporate context from both sequence directions [5]. Transformer models provide an alternative contextual representation [6]. BERT demonstrated the effectiveness of large-scale language-model pretraining followed by task-specific fine-tuning [7]. Sahmoud and Mikki applied BERT to spam detection [8]. XLM-R extends multilingual pretraining across many languages, including Bangla, making it a relevant candidate for multilingual text classification [9].

Predictive scores alone do not show which parts of a message influenced an individual decision. LIME approximates a model locally by perturbing an input and fitting an interpretable surrogate model [10]. Table I summarizes representative methods and their relevance to this bilingual setting.

III. GAP ANALYSIS

The reviewed work motivates four specific gaps addressed by this experiment. First, English benchmarks and English-focused classifiers dominate the cited SMS spam literature, leaving Bangla and code-mixed evaluation underrepresented. Second, aggregate scores can conceal performance differences between a high-resource language and a smaller language subset. Third, model comparisons made on different datasets or partitions do not isolate the effect of multilingual pretraining; a shared split provides a more direct, though still limited,

TABLE I
COMPARISON OF REPRESENTATIVE SMS SPAM DETECTION APPROACHES.

Work	Approach	Language scope	Limitation for this study
Almeida et al. [1]	Classical ML benchmark	English	No multilingual model or Bangla evaluation
Roy et al. [2]	CNN, LSTM, and classical ML	English	No English–Bangla comparison or token-level explanation
Abayomi-Alli et al. [3]	BiLSTM and other models	Local SMS data	Does not isolate multilingual transfer in this setting
Sahmoud and Mikki [8]	Fine-tuned BERT	English-focused	Does not evaluate Bangla alongside English
This study	BiLSTM, XLM-R, and LIME	English and Bangla	Bangla test set is small ($n = 86$)

TABLE II
CLASS COMPOSITION OF THE CLEANED DATASET.

Language	Ham	Spam	Total	Spam rate
English	4,516	642	5,158	12.4%
Bangla	276	199	475	41.9%
Total	4,792	841	5,633	14.9%

comparison. Fourth, standard metrics do not reveal whether a classifier responds to plausible spam cues or incidental tokens, and local explanations for Bangla examples are limited in this study’s reference set.

Accordingly, this work evaluates BiLSTM and XLM-R on one bilingual split, reports XLM-R scores separately for English and Bangla, and uses LIME on selected English and Bangla messages. This is a scoped empirical comparison, not evidence that the broader literature lacks multilingual SMS research or that the proposed system is novel in general.

IV. METHODOLOGY

A. Dataset

The English messages originate from the SMS Spam Collection [1]. The Bangla source file contained 503 messages (216 spam and 287 ham). After combining the sources and removing duplicate and invalid entries, the reported dataset contains 5,633 messages: 5,158 English and 475 Bangla. Each record has message text, a ham/spam label (0/1), and a language tag. Table II gives the final class counts. The class distribution is imbalanced overall: spam represents 14.9% of the complete dataset, while 41.9% of the Bangla subset is spam.

Figure 1 summarizes the reported workflow, and Figure 2 visualizes the class distribution. The data were divided into 4,506 training messages (3,833 ham and 673 spam) and 1,127 test messages (959 ham and 168 spam), using an 80/20 split intended to preserve the overall spam rate. The test set contains 1,041 English messages and 86 Bangla messages; the Bangla test subset includes 48 ham and 38 spam messages. The source paper does not specify a split random seed or fully describe the stratification procedure.

B. Text preprocessing

Text columns were converted to string values and the label column was excluded from model inputs. For BiLSTM, a

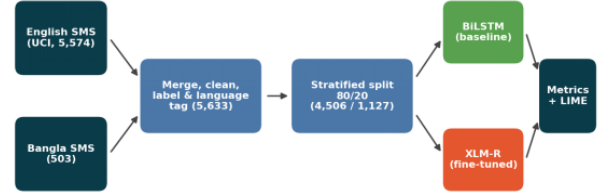


Fig. 1. Overall workflow for bilingual SMS spam detection.

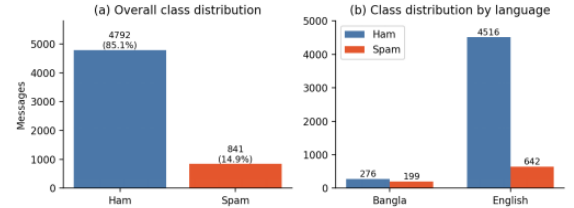


Fig. 2. Ham and spam message counts in the complete dataset and by language.

Keras tokenizer with a 10,000-word vocabulary and an out-of-vocabulary token converted messages to integer sequences. Sequences were padded or truncated to 100 tokens. XLM-R used its SentencePiece tokenizer, with a model limit of up to 512 tokens. No language-specific stemming or stop-word removal was applied; numbers, codes, and punctuation were retained because they may be informative for spam classification.

C. Model architectures

The BiLSTM baseline consists of an embedding layer, a bidirectional LSTM layer with 64 units, dropout with rate 0.5, and a sigmoid dense output for the spam probability. The bidirectional layer processes each sequence in both directions [4], [5].

XLM-R is a Transformer encoder with 12 layers and a hidden size of 768. A sequence-classification head was added, and all model weights were fine-tuned on the spam task. The multilingual pretrained representation is intended to support both English and Bangla text [9].

D. Training and evaluation

The BiLSTM used the Adam optimizer and binary cross-entropy loss for five epochs, with batch size 32 and 10% of the

training data reserved for validation. XLM-R was fine-tuned on the same training split and evaluated on the same test split. The available paper does not report several settings needed to reproduce the transformer run, including its learning rate, batch size, epoch count, maximum sequence length used in practice, and random seed.

Because the classes are imbalanced, the study reports accuracy, precision, recall, and F1-score for the spam class, along with confusion counts in the order true negative (TN), false positive (FP), false negative (FN), and true positive (TP):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (3)$$

$$F_1 = \frac{2 \text{ Precision Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

LIME perturbs tokens in an individual message and fits a local interpretable model to estimate token contributions [10]. The reported analysis covers one English spam message, one English ham message, and one Bangla spam message.

E. Reproducibility information

The paper identifies the dataset sources and the BiLSTM settings above, but does not provide a dependency lock-file, split seed, complete XLM-R training configuration, or model-weight download instructions. The project repository is available at <https://github.com/sabbirOG/ML-Report262>. A `requirements.txt` file, source code, and trained weights (or download instructions) should accompany the repository. These missing details limit exact reproduction of the reported results.

V. RESULTS AND DISCUSSION

A. Overall test-set performance

Table III reports the metrics on all 1,127 test messages. XLM-R has a higher accuracy by 0.36 percentage points and a higher F1-score by 1.58 points. Its recall is 6.55 points higher, while its precision is 3.64 points lower than BiLSTM. The confusion matrices in Figure 3 show the operational trade-off: XLM-R reduces false negatives from 19 to 8, but increases false positives from 6 to 13. Figure 4 compares the four reported metrics. Thus, the higher recall may be useful where missed spam is costly, but the increased false-positive count may burden users with more incorrectly flagged messages. Accuracy alone understates this trade-off because ham messages form most of the test set.

B. Performance by language

Table IV and Figure 5 present the XLM-R results by language. English performance is higher across the reported metrics: the English subset has 98.94% accuracy and 95.79% F1-score, compared with 88.37% accuracy and 87.50% F1-score for Bangla. Bangla recall remains high at 92.11%,

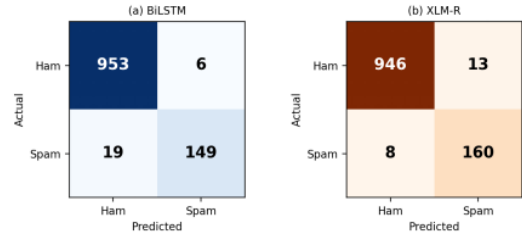


Fig. 3. Confusion matrices for BiLSTM and XLM-R on the test set. Rows are actual classes and columns are predicted classes.

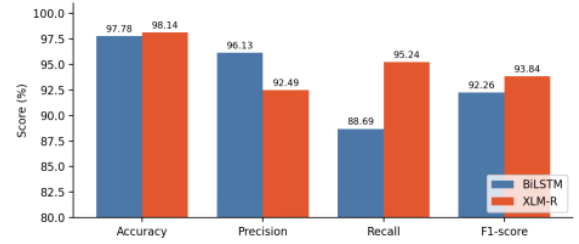


Fig. 4. Comparison of accuracy, precision, recall, and F1-score for both models.

but precision is 83.33%, indicating more false alarms among Bangla messages. The Bangla test set has only 86 examples, so individual errors substantially affect its percentages; these estimates should be treated as preliminary rather than as stable language-level performance.

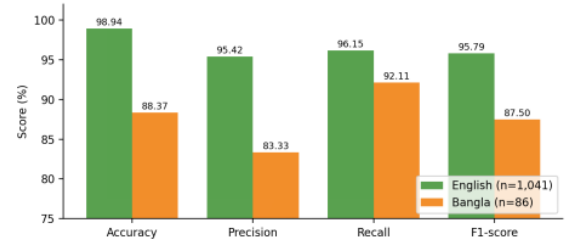


Fig. 5. XLM-R accuracy, precision, recall, and F1-score for English and Bangla test messages.

C. LIME-based interpretation

The LIME examples provide local, instance-specific evidence rather than a global account of model behavior. For an English spam message, XLM-R assigned a spam probability of 99.82%. The strongest positive token contributions shown in Figure 6 are a campaign-like code and a phone number, followed by numeric and promotional terms. For an English ham example, the prediction was near the decision boundary (50.77% ham and 49.23% spam); words associated with ordinary language supported the ham prediction, while terms such as “SMS” and capitalized words contributed toward spam. For the Bangla spam example, the spam probability was 99.63%; a USSD-like code and a price-per-day expression were among the strongest positive contributions. These examples suggest

TABLE III

TEST-SET PERFORMANCE FOR THE SPAM CLASS ($n = 1,127$). METRIC DIFFERENCES ARE PERCENTAGE POINTS; FN AND FP DIFFERENCES ARE COUNTS.

Model	Accuracy	Precision	Recall	F1-score	FN / FP
BiLSTM	97.78%	96.13%	88.69%	92.26%	19 / 6
XLm-R	98.14%	92.49%	95.24%	93.84%	8 / 13
Difference (XLm-R – BiLSTM)	+0.36	−3.64	+6.55	+1.58	−11 / +7

TABLE IV
XLm-R TEST-SET PERFORMANCE BY LANGUAGE.

Subset	Accuracy	Precision	Recall	F1-score	n
English	98.94%	95.42%	96.15%	95.79%	1,041
Bangla	88.37%	83.33%	92.11%	87.50%	86
Overall	98.14%	92.49%	95.24%	93.84%	1,127

that numerical patterns and promotional language influence some predictions, but three selected messages cannot establish explanation reliability or broad model behavior.

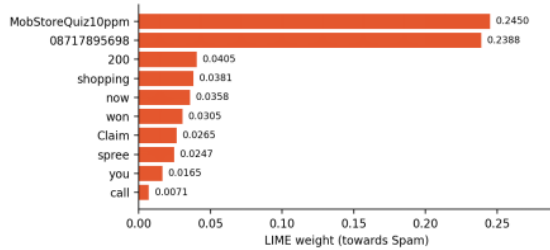


Fig. 6. LIME token contributions toward the spam prediction for one English spam example (predicted spam probability: 99.82%).

VI. CONCLUSION

This study compared BiLSTM and fine-tuned XLm-R for spam classification on a cleaned English–Bangla SMS dataset and used LIME to inspect selected predictions. XLm-R achieved a higher test F1-score and recall, and missed fewer spam messages, but generated more false positives than BiLSTM. Performance was lower for Bangla than for English, while the Bangla recall remained relatively high. Because the Bangla test subset contains just 86 messages and the evaluation uses one split without repeated runs or confidence intervals, the results do not establish robust Bangla performance or generalization to other populations.

Future work should collect and label more Bangla and code-mixed messages, use repeated or cross-validated evaluation, document all training and data-splitting settings, and study threshold choices and error cases. Broader comparisons with multilingual pretrained models and more extensive human-reviewed explanation analysis would also strengthen the evidence. Before submission, the authors should complete the student IDs and section, verify dataset provenance and permissions, and add the required repository and reproduction instructions. The reported experiment is a research prototype, not a deployed spam-filtering service.

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